

SOFTWARE REQUIREMENTS SPECIFICATION

Ranjani AI Chat Application

Project Type: AI-based Chat Application

Document Version: 1.0

Prepared for Academic Project Documentation

1. Introduction

Ranjani AI Chat Application is a web-based chat application designed to provide a simple and user-friendly interface for interacting with an AI assistant. The application follows a familiar messaging-style interface and includes features such as chat messages, AI typing indication, search, dark mode, image upload, emoji support, voice input interface, message status ticks, and chat management.

2. Purpose

The purpose of this system is to provide an easy-to-use platform where a user can enter questions or messages and receive AI-generated responses through a conversational interface. The project also demonstrates the use of front-end web technologies to build an interactive AI chat experience.

3. Scope

- Provide a messaging interface between the user and an AI assistant.
- Display user messages and AI responses in a chat window.
- Show an AI typing indicator while a response is being prepared.
- Provide search, dark mode, image upload, emoji and voice-input interface features.
- Provide message status indicators and basic chat management.
- Allow the application to be extended with a real AI/API backend in the future.

4. Overall Description

The application is intended for users who want a simple conversational interface. The front end is implemented as a web application using HTML, CSS and JavaScript. The interface is designed to work on common mobile and desktop browsers. The current project focuses mainly on the user interface and interaction flow, while a production deployment can connect the application to an AI service through a secure backend.

5. Functional Requirements

ID	Requirement	Description
FR-01	User Interface	The system shall display a chat interface with header, message area and message input area.
FR-02	Send Message	The system shall allow the user to enter and send a message.
FR-03	AI Response	The system shall display an AI response for a submitted user message.
FR-04	Typing Indicator	The system shall show an AI typing indicator while preparing a response.
FR-05	Search	The system shall provide a chat search facility.
FR-06	Dark Mode	The system shall allow the user to switch between normal and dark themes.
FR-07	Image Upload	The system shall provide an option to select and send an image.

ID	Requirement	Description
FR-08	Emoji	The system shall provide an emoji input option.
FR-09	Voice Input	The interface shall provide a voice-input control.
FR-10	Message Status	The system shall display message delivery/read-style tick indicators.
FR-11	Chat Management	The system shall provide options for clearing or deleting chat content.
FR-12	Name and Time	The chat interface shall display relevant sender/name and time information.

6. Non-Functional Requirements

- **Usability:** The interface should be simple and understandable for a beginner user.
- **Performance:** User interactions should respond without unnecessary delay.
- **Reliability:** The application should handle normal user interactions without unexpected page failures.
- **Maintainability:** HTML, CSS and JavaScript should remain separated so features can be updated easily.
- **Compatibility:** The application should work on modern web browsers and common mobile screen sizes.
- **Security:** Any future AI/API credentials should be kept on a secure backend and not exposed in client-side code.

7. System Requirements

Software: Modern web browser, HTML5, CSS3 and JavaScript-enabled environment.

Hardware: Smartphone, tablet or computer capable of running a modern web browser.

Development: The project can be edited using a mobile code editor or a desktop development environment.

8. User Interface Requirements

- Application header containing the application name and controls.
- Chat area for displaying conversation messages.
- Input area for typing and sending messages.
- Search control for finding chat content.
- Theme control for dark mode.
- Controls for emoji, image and voice input.
- Typing indicator and message status indicators.

9. Data and Storage Requirements

The application may store chat information locally when local storage or another client-side storage mechanism is implemented. A future backend may store user sessions and conversations using an appropriate database. Sensitive data should not be stored unnecessarily.

10. Future Enhancements

- Connect the front end to a secure AI API through a backend service.
- Add user authentication and account management.
- Add persistent cloud chat history.
- Improve voice input and speech output.
- Add richer file and document handling.
- Add message editing, copying and sharing features.
- Improve responsive design and accessibility.

11. Conclusion

Ranjani AI Chat Application provides a practical web-based interface for AI-assisted conversation. The SRS defines the main functional and non-functional requirements, system environment, interface expectations and possible future enhancements. The document can be used as part of the project's academic documentation along with the UML diagrams and source code.